

1) At the isoelectric PH of an amion acid which has two pka values the net charge is :

1) -1

2) 0

3) 1

4) 0.5

2) which of the following amino acid is the most soluble in water ?

1) glycine

2) tryptophan

3) tyrosine

4) phenylalanine

3) A solution with a concentration of 0.18 M was found to have an absorbance of 0.42 . Another solution of the same sample is measured under the same condition and was found to have an absorbance of 0.34 . what is the concentration ?

1) 0.062

2) 0.146

3) 0.0429

4) 0.18

4) The material that used to staining protein in electrophoresis is :

- 1) Bromophenol blue
- 2) methyl orange
- 3) ninhydrin
- 4) comassie brilliant blue

5) At PH= 10 , Addition of ninhydrin solution to glutamine followed by boiling will give :

- 1) no reaction
- 2) yellow color
- 3) brown color
- 4) brown color

6) which of the following amino acid does not have hydroxyl group in side chain ?

- 1) tyrosine
- 2) threonine
- 3) isoleucine
- 4) serine

7) which of the following substance is used in electrophoresis ?

- 1) cellulose acetate
- 2) sodium dodecyl sulfate

3) polyacrylamide gel

4) All of them

8) for 2M protein sample studied by biuret test . the absorption of the blank is ?

1) zero

2) 2

3) not predictable

4) 4

9) FOR ARGININE $PI = 10.76$ $pka1 = 2.18$ & $pka2 = 8.32$ what is the value of $pka3$?

1) 13.2

2) 8032

3) 3.45

4) 2.09

10) suppose you have amixture of 2 dyes (the maximum wavelength 1 = 460 nm , the maximum wavelength 2 = 465 nm) the relation between the concentration of the two dyes 1 and 2 is ?

1) $c1 > c2$

2) $c1 < c2$

3) $c1 = c2$

4) can not be determined

11) amino acid has a $R_f = 0.25$ how far will x traveled from the origin when eluent front has traveled 8 cm ?

- 1) 4 cm
- 2) 0.063 cm
- 3) 2 cm
- 4) 32 cm

12) in electrophoresis , the buffer choice depend on ??

- 1) R-group
- 2) PI
- 3) size of protein
- 4) polarity of amino acid

13) the pka of cystein side chain ($\text{CH}_2\text{-SH}$) group is 8.03 so it is about 80% dissociated at PH ?

- 1) 8.4
- 2) 8.0
- 3) 8.9
- 4) 8.8

14) which amino acid reacts with ninhydrin to give purple complex and give positive result in xanthoproteic reaction ?

- 1) alanine
- 2) proline
- 3) tyrosin

4) glycine

15) which of these cause complete irreversible denaturation of protein ?

1) 100 c

2) -10 c

3) all of them

4) alkaline solution

Blaze Batch 😊